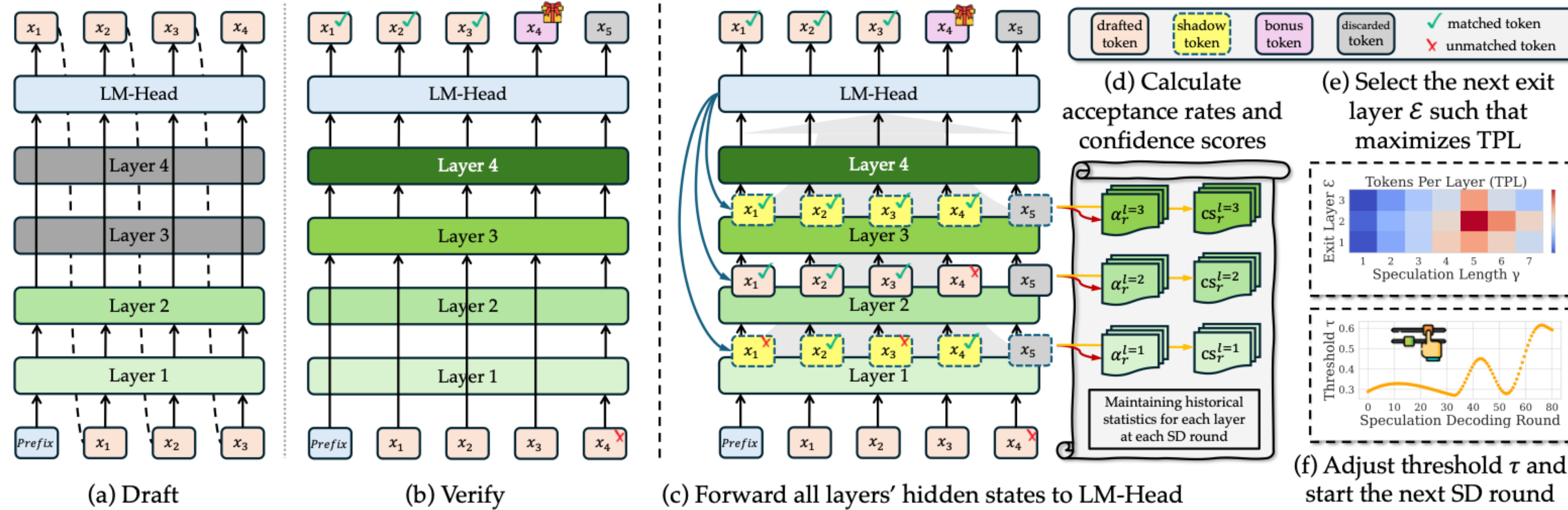




Abstract

- LLM inference is **slow** due to auto-regressive decoding and memory **bandwidth**-bound layer loads.
- Speculative Decoding (SD) drafts γ tokens with a cheaper model, then verifies in parallel with the target LLM.
- LayerSkip (LS) enables self-speculation: use first $\mathcal{E}$ layers to draft, full model to verify, saving memory and compute.
- However, the optimal **exit layer $\mathcal{E}$** and **speculation length γ** vary across tasks and even within a sequence.
- DEL** reuses intermediate hidden states to **adapt $\mathcal{E}$ and γ** in real-time during inference.

Methodology

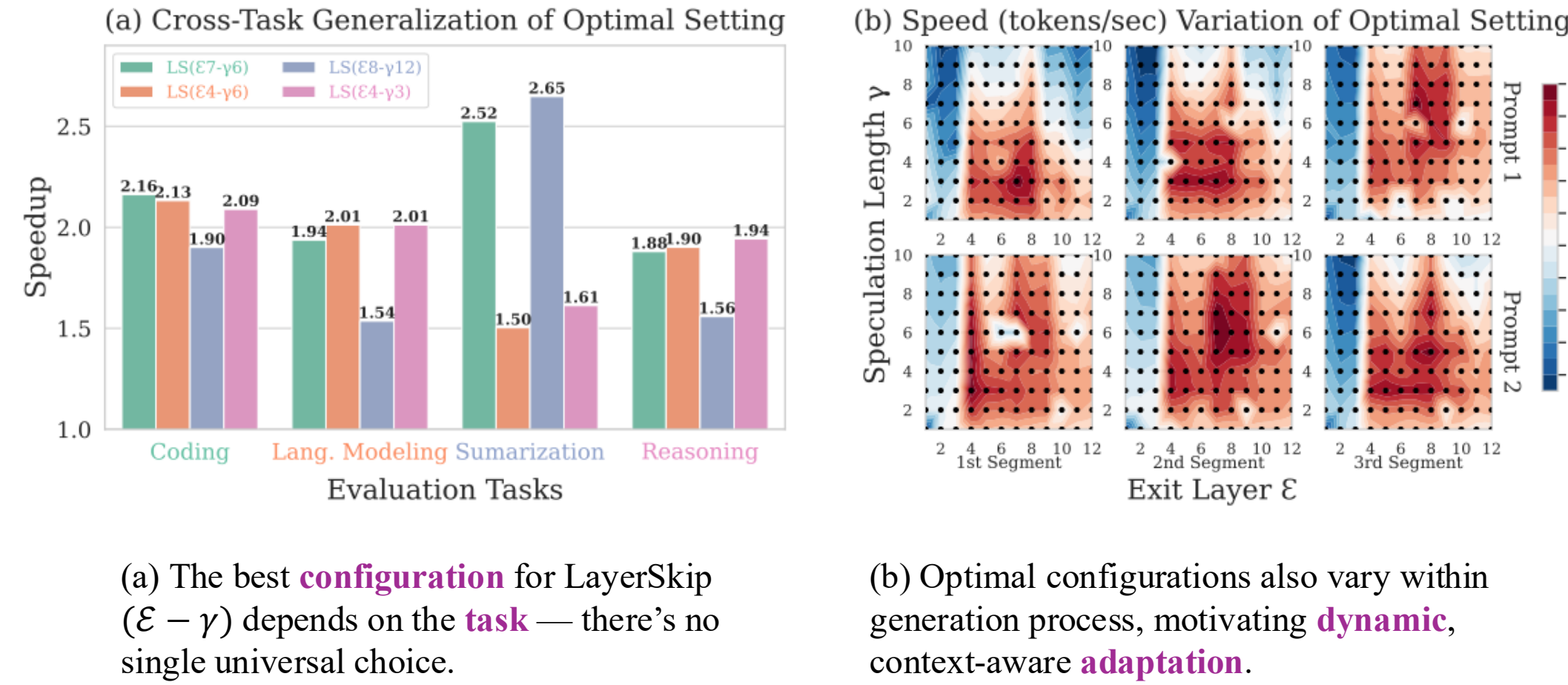


(a–b) In this LayerSkip example, Layer 2 serves as the exit layer, drafting **4 tokens** auto-regressively. During verification, **3 accepted**, while x_4 is rejected and becomes the bonus token.

(c–d) The plug-and-play DEL module reuses the **hidden states** from all layers, forwarding them through the LM-Head to generate **shadow tokens** used to estimate layer-wise token **acceptance rates** and **confidence scores** without incurring additional forward passes.

(e–f) DEL selects the next exit layer $\mathcal{E}$ that maximizes **Token-per-Layer (TPL)** based on historical acceptance statistics, and updates the dynamic confidence threshold τ to guide adaptive speculation length γ in the next SD round.

Observations

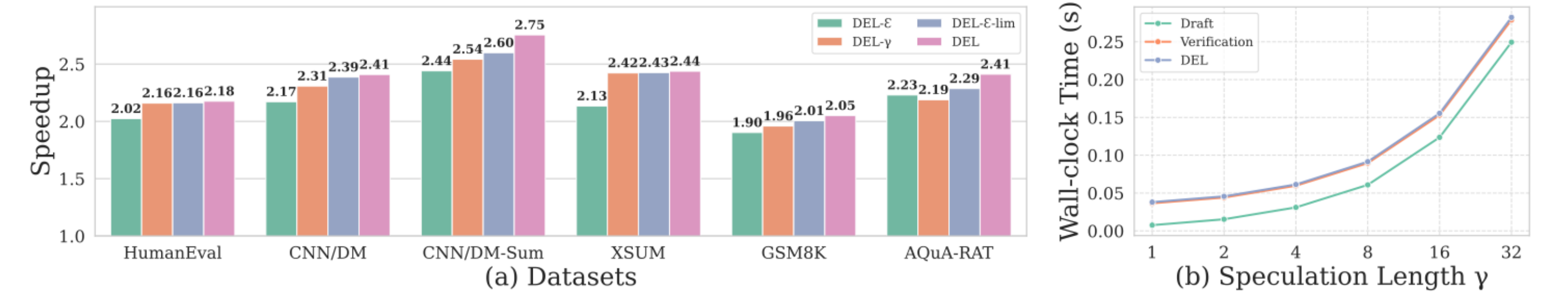


(a) The best **configuration** for LayerSkip ($\mathcal{E} - \gamma$) depends on the **task** — there's no single universal choice.

(b) Optimal configurations also vary within generation process, motivating **dynamic**, context-aware **adaptation**.

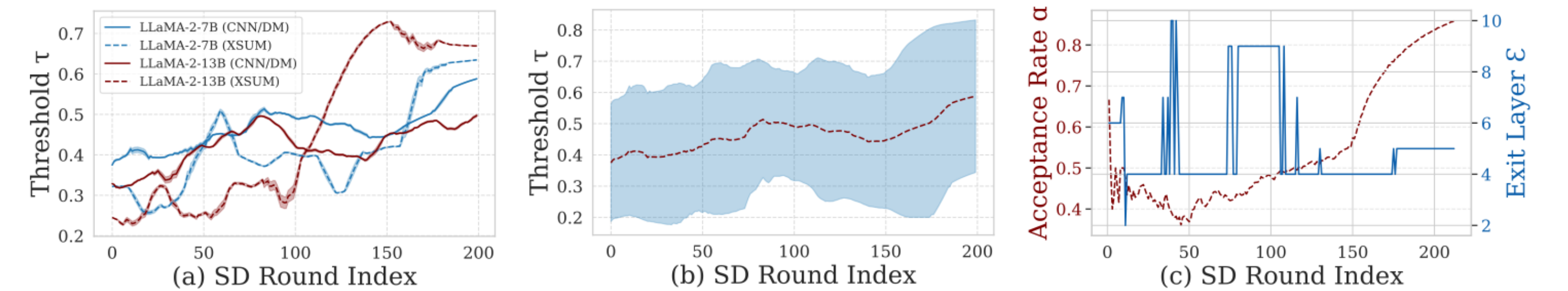
Results

Models	Methods	AQuA-RAT (Reasoning)		CNN/DM (Lang. Mod.)		CNN/DM (Abs. Sum.)		XSUM (Abs. Sum.)		Speed (tokens/s)	Overall Speedup
		eTPL	Speedup	eTPL	Speedup	eTPL	Speedup	eTPL	Speedup		
LLaMA -3-8B	Vanilla	0.031	1.00×	0.031	1.00×	0.031	1.00×	0.031	1.00×	32.57	1.00×
	LS($\mathcal{E}3-\gamma6$)	0.084	2.17×	0.087	2.29×	0.089	2.30×	0.077	2.08×	72.00	2.21×
	FS($\mathcal{E}3-\gamma6$)	0.094	2.47×	0.102	2.76×	0.087	2.30×	0.080	2.18×	79.13	2.43×
	DV($\mathcal{E}3$)	0.052	1.52×	0.052	1.50×	0.085	2.32×	0.051	1.46×	55.40	1.70×
	DEL	0.098	2.53×	0.109	2.81×	0.105	2.72×	0.092	2.40×	85.20	2.62×
LLaMA -2-70B	Vanilla	0.013	1.00×	0.013	1.00×	0.013	1.00×	0.013	1.00×	9.75	1.00×
	LS($\mathcal{E}9-\gamma6$)	0.033	2.51×	0.026	1.98×	0.028	2.11×	0.026	1.99×	20.90	2.14×
	FS($\mathcal{E}9-\gamma6$)	0.036	2.76×	0.027	2.09×	0.029	2.22×	0.026	2.00×	22.08	2.26×
	DV($\mathcal{E}9$)	0.021	1.59×	0.019	1.50×	0.020	1.51×	0.019	1.51×	14.86	1.52×
	DEL	0.038	2.84×	0.028	2.15×	0.035	2.57×	0.028	2.12×	23.49	2.41×



(a) Ablation on LLaMA2-7B shows both dynamic exit layer selection and dynamic draft-exiting components improve performance.

(b) Runtime breakdown across γ shows DEL adds minimal overhead.



(a) Confidence threshold τ evolves over 200 SD rounds across models and datasets.

(b) τ for a sample prompt, with accepted (top) and rejected (bottom) token scores shown as a shaded range.

(c) Evolution of selected exit layer $\mathcal{E}$ and acceptance rate α during inference.